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RS-YOLO: A Lightweight Improved Framework of YOLOv12 for Multi-Target Recognition in Remote Sensing Images

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***Abstract*—In high-resolution remote sensing images, the significant scale differences among targets, complex backgrounds, and dense distribution of small targets pose severe challenges to traditional detection models. To address this, this paper proposes RS-YOLO, a lightweight multi-target detection model specifically designed for remote sensing scenarios. Firstly, by optimizing the backbone network architecture, we introduced and enhanced the Dynamic Tanh (DyT) normalization mechanism. Given the characteristic of extreme value density in remote sensing images, DyT achieves activation scaling through dynamic learning rate adjustment and channel-independent parameter design, thereby avoiding the statistical dependency issues associated with traditional normalization. Secondly, we optimized and introduced an enhanced IoU loss function (Inner-IoU), which integrates center prior, rotation angle weighting factors, and small target weighting mechanisms to weight-correct IoU. Experimental results on the DOTA-v1.0 dataset show that RS-YOLO achieves 78.9% mAP_{50} on DOTA with only 2.52M parameters and 5.9G floating-point operations (FLOPs), and an inference latency of 35.8 milliseconds, fully demonstrating its lightweight advantage.Notably, the model's mAP_{50} exceeds 80% in the detection tasks of multiple typical targets. Significant performance improvements are particularly evident in difficult cases such as extremely small targets and high-rotation targets. RS-YOLO provides an efficient and feasible solution for rapid multi-target inspection in remote sensing images.**

Keywords—Remote sensing imagery; Multi-object recognition; Deep learning; YOLOv12; Loss function.

I. INTRODUCTION

In recent years, remote sensing technology has developed rapidly, with its applications becoming increasingly diverse across various fields, playing a significant role in both military and civilian sectors[1-2]. It serves as a crucial foundation for applications such as urban planning, land use, and natural disaster monitoring. Remote sensing image target detection is an important research direction in the field of computer vision, aiming to detect and identify specific targets from remote sensing images[3-4]. Given this, enhancing the performance of remote sensing target detection remains a critical research direction.

In traditional feature-based object detection methods, the process involves first extracting candidate regions, then designing manual features, and finally using a classifier for detection[5]. However, remote sensing images have high resolution and large data volumes, and manual feature descriptors have limitations in representing diverse features, which prevents the algorithm from fully exploiting deep semantic information[6]. Therefore, breaking through the limitations of traditional methods and expanding the application of deep learning-based remote sensing image target detection technology has significant research value and application prospects[7-8].

In the field of remote sensing multi-object detection, deep learning-based algorithms have made significant progress. However, existing research primarily relies on earlier frameworks such as YOLOv5[9], YOLOv8[10], and YOLOv11[11]. As the latest version, YOLOv12[12] has certain advantages over earlier versions in remote sensing scenarios. However, YOLOv12 also has some shortcomings. First, its normalization layer involves redundant computations, requiring real-time calculation of the mean and variance of the input, which consumes a significant amount of computational resources in long sequence tasks and increases model inference latency. Second, the IoU loss function in YOLOv12 has poor adaptability to rotated targets, making it difficult to effectively handle common issues such as target rotation and aspect ratio changes in remote sensing images. Given that there is still room for improvement in model lightweighting and detection accuracy, YOLOv12 was ultimately selected as the baseline model, with the following specific improvements:

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DyT Normalization Mechanism[13]: In this study, we introduced a DyT normalization mechanism to replace the traditional normalization layer in the A2C2f module, thereby constructing a lightweight backbone network module called A2DyT. This replacement can effectively reduce the number of network parameters and computational complexity, thereby lowering the model's storage and computational requirements.

Inner-IoU Loss Function[14]: In the context of detecting objects within remote sensing images, we have introduced an innovative inner intersection-over-union (Inner-IoU) loss function. This novel approach replaces the conventional YOLOv12 loss function, offering a superior balance between classification and localization accuracy. Additionally, it accelerates model convergence speed, improves training efficiency, and enables the model to achieve higher detection performance within a limited training time.

II. Methodology

A. Backbone Network Optimization with DyT Mechanism

In this study, we improved upon YOLOv12 and proposed a deep learning model for multi-object detection in remote sensing images (as shown in Fig.1) called RS-YOLO. In recent years, the Transformer architecture has been widely adopted in computer vision due to its powerful modeling capabilities, achieving significant improvements across various tasks. Motivated by these successes, transferring key Transformer modules to object detection holds considerable research value. Among these, normalization layers serve as core components within the Transformer architecture, playing an indispensable role in stabilizing model training and enhancing feature representation. The general computational formulas for traditional normalization layers, such as Layer Normalization (Layer Norm) and Root Mean Square Normalization (RMS Norm), are defined as follows:

$$\text{normalization}(x) = \gamma * \left(\frac{x-\mu}{\sqrt{\sigma^2+\epsilon}}\right) + \beta \tag{1}$$

Here, the input tensor is denoted as $x \in R^{B\times T\times C}$, where B , T and C represent the batch size, sequence length (or spatial dimensions), and feature channels, respectively. The terms μ and σ^2 denote the mean and variance of the input along the normalization dimension(s). A small constant ϵ is added to prevent numerical instability during division. Furthermore, γ and β are learnable, channel-wise affine parameters responsible for scaling and shifting the normalized values.

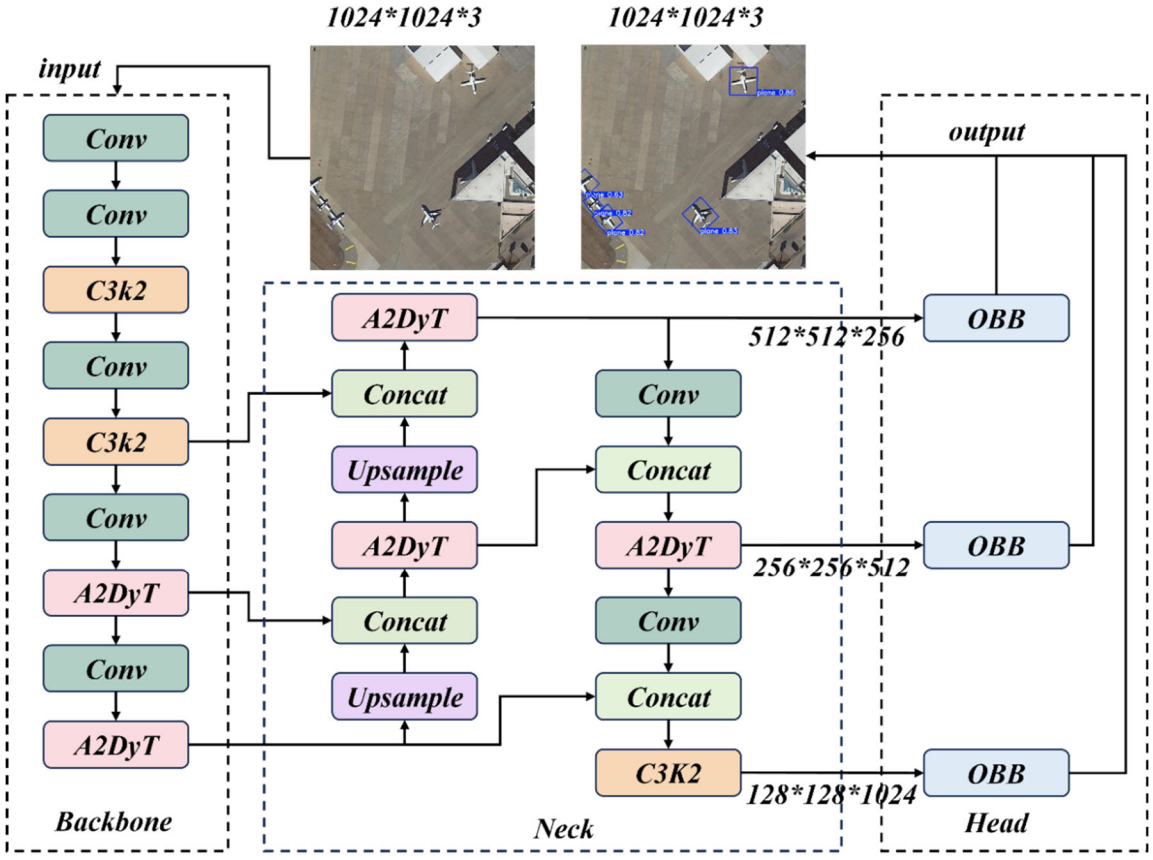


Fig. 1. Network architecture of the RS-YOLO

As illustrated in Fig.2(a), traditional normalization layers within the Transformer framework suffer from two primary limitations.

1)High Computational Cost: The requirement for real-time computation of input statistics (μ and σ^2) consumes significant computational resources, particularly for tasks involving long sequences. This overhead substantially increases model inference latency.

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2)Structural Complexity: The prevalent "Add-and-Norm" module, interleaved between Transformer layers, introduces additional parameters and computational steps. This design increases overall model complexity, complicates debugging, and impedes subsequent optimization efforts.

To mitigate these limitations, we propose DyT normalization mechanism, a novel normalization mechanism (depicted in Fig.2(b)). DyT reconstructs the normalization process at the element level, retaining the core functions of traditional normalization while achieving superior performance through structural innovation. Its core formulation is defined as follows:

$$DyT(x) = \gamma * \tanh(\alpha x) + \beta \qquad (2)$$

Within this formulation, x denotes the input tensor. The parameter α is a learnable scalar that dynamically scales the input range. The hyperbolic tangent function (tanh) is applied to compress extreme values, mapping the input to the interval [-1, 1]. The parameters γ and β are learnable, channel-wise affine vectors, responsible for the final scaling and shifting operations.

Crucially, DyT achieves precise compression of extreme values, thereby mitigating background noise interference and enhancing feature stability—all without incurring additional computational overhead. In the present work, we integrate DyT into the backbone network of YOLOv12; specifically, we replace the original A2C2f modules with newly designed A2DyT modules, as illustrated in Fig.2(c).

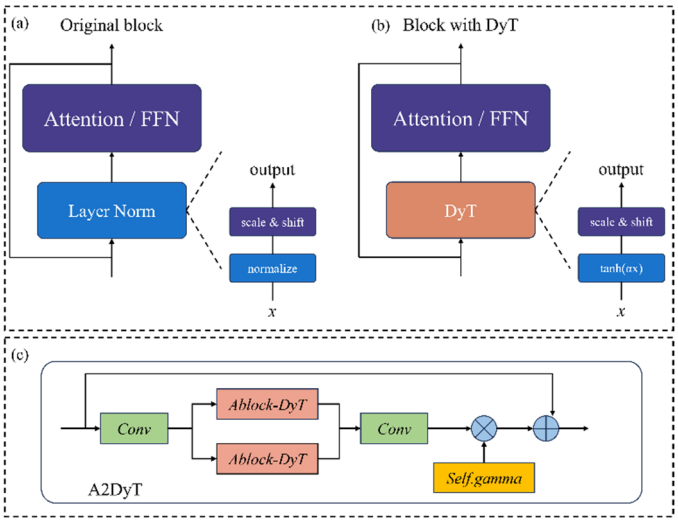


Fig. 2. (a) Original Transformer Block with Layer Normalization. (b) Transformer Block with Proposed Dynamic Tanh (DyT) Normalization Mechanism.(c) A2DyT Block: Architecture Integrating DyT into Dual-Path Convolutional Branches

In this study, to address the issue of feature distortion caused by non-stationary feature distributions resulting from significant grayscale differences between targets and backgrounds in remote sensing images, we introduce the DyT normalization mechanism with specific optimizations. This overcomes the limitations of traditional normalization methods that rely on global statistics. The optimizations are primarily manifested in two aspects: Firstly, the initial learning rate for the parameter α, which controls the activation value range, is set to 0.1 and dynamically adjusted using a cosine decay scheduling strategy. This enables the compression intensity of the tanh function for extreme values to adaptively evolve throughout the training process. Specifically, it increases compression intensity in high-contrast regions (e.g., bright metallic targets like plane and large vehicle) to suppress noise, while reducing compression in low-contrast regions (e.g., indistinct small structures like harbors and storage tanks) to preserve detail. Secondly, the shared scalar parameter α used in the original work [13] is improved into channel-wise learnable vectors. This design aims to distinguish the feature distribution disparities and dynamic ranges across different semantic channels within remote sensing images (e.g., edge channels versus texture channels), thereby enabling more precise channel-level activation scaling.

Through these optimizations, the DyT Normalization mechanism significantly reduces computational costs while enhancing adaptability to the non-stationary feature distributions prevalent in remote sensing imagery, effectively mitigating feature distortion.

B. Loss Function Improvement via Inner-IoU Design

In the task of multi-object and multi-scene detection in remote sensing images, YOLOv12 adopts a total loss function that is a weighted sum of localization loss, confidence loss, and classification loss. However, due to the characteristics of remote sensing ground objects such as significant scale differences, complex and variable backgrounds, and extreme class imbalance, traditional fixed weights fail to adaptively characterize the dynamic contributions of various losses, thereby limiting the further improvement of detection accuracy. To address this issue, this study innovatively introduces the Inner-IoU[14] bounding box regression method (as shown in Fig.3).

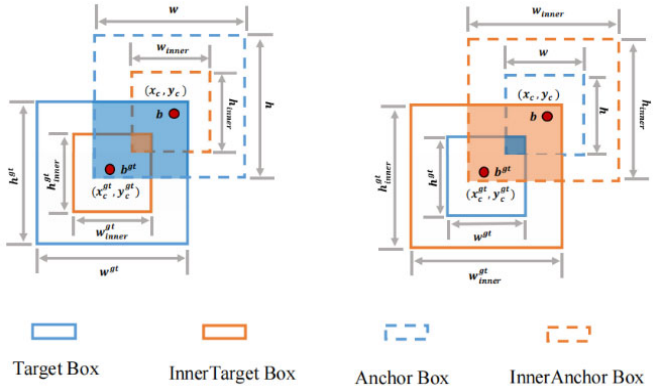


Fig. 3. The architecture of Inner-IoU

Firstly, we introduce a center-weighted mechanism, which emphasizes the weight of the distance between the predicted bounding box and the true bounding box center, thereby focusing the loss calculation on the alignment of the target's geometric center. This improvement is crucial for detecting rotated targets in remote sensing images, as the bounding box offset of rotated targets often accompanies subtle deviations in the center position, and center alignment can effectively constrain the regression accuracy of the rotation angle. Secondly, for small targets with an area less than 32×32 pixels, the IoU calculation weight is increased, and a rotation angle

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difference factor is introduced (increasing the correction strength when the angle deviation exceeds 30°). This ensures that the model can accurately learn the bounding box features of small and rotating targets even in complex backgrounds. This improvement not only enables bounding box regression to adapt to common characteristics of remote sensing targets, such as rotation and drastic changes in aspect ratio, but also specifically alleviates the localization ambiguity issues of small and rotating targets. Ultimately, by providing more precise supervision signals, the model's robustness and detection accuracy in remote sensing scenarios are significantly enhanced.

III. Experiment

A. Experiment Dataset

To ensure the reproducibility and impartiality of the experiment, we used the publicly available authoritative dataset DOTA-v1.0[15]. (available at:https://captain-whu.github.io/DOTA/dataset.html). The original dataset comprises 2,806 images of variable dimensions, containing 188,282 annotated instances spanning 15 typical aerial object categories: plane , ship, storage tank , baseball diamond , tennis court , basketball court , ground track field , harbor , bridge , large vehicle , small vehicle , helicopter, roundabout, soccer ball field , and swimming pool. The fully annotated dataset was first partitioned into training and validation subsets at an 8:2 ratio. Subsequently, the images were uniformly cropped into non-overlapping patches of 1,024 × 1,024 pixels using a sliding window approach. This process yielded 15,749 training patches and 5,297 validation patches for model training and evaluation.

B. Environment and Parameter Configuration

During the experimental training phase of this study, all tasks were performed on a high-performance workstation equipped with the Windows 11 operating system, an Intel Core i7-14700k CPU, and an NVIDIA GeForce RTX 3090 GPU (24GB VRAM). For the deep learning framework, Python 3.9.0 and PyTorch 2.4.1 were used for model development and training, with CUDA 12.1 and CUDNN 8.9.7 implemented to enable efficient GPU-accelerated training, fully leveraging hardware capabilities to ensure the efficiency and stability of model training. To explore and optimize the performance of the YOLOv12 algorithm, this study selected YOLOv12n as the baseline reference model. Table I shows the configuration details of key parameters during training, while the remaining parameters follow the default settings of YOLOv12n to ensure standardized and comparable experimental conditions, laying the foundation for subsequent algorithm improvement research.

TABLE I. Model hyperparameter settings

Parameter Name	Value
Epochs	300
Batch Size	32
Input Image Size	1024 x 1024
Optimizer	Adam
Learning Rate	0.001
Initial Momentum	0.9
Bounding Box Loss Weight	7.5
Classification Loss Weight	0.5
Mosaic Data Augmentation Probability	1.0

C. Indicators for evaluation

To comprehensively evaluate the model's performance in object detection tasks, multiple key metrics were employed. These metrics include Precision, Recall, mAP_{50}, and mAP_{50-95}, which collectively serve as quantitative indicators for assessing the detection performance of SOTA models in this study. In addition to these performance-related metrics, this paper also uses the number of network model parameters (#Param.), the number of floating-point operations (Flops), and inference latency (lat.) as critical metrics for evaluating model efficiency.

D. Analysis of testing results

In this study, we selected multiple YOLO series models for comparative analysis. As shown in Table II, RS-YOLO has achieved remarkable superiority over other object detection algorithms in defect detection performance. This model boasts a compact structure, with only 2.52 M parameters and a mere 5.9G Flops of floating-point operations. Moreover, it achieves an inference latency of just 35.8 milliseconds, which is the lowest among all the algorithms compared. This highlights its outstanding efficiency. When compared to the baseline model YOLOv12, RS-YOLO has managed to reduce the number of parameters by 1.5%, the volume of floating-point operations by 8.4%, and the inference latency by 3.7%.

TABLE II. Comparison of defect detection results of different object detection algorithms

Methods	#Param. (M)	Flops (G)	Lat. (ms)	Precision %	Recall %	mAP_{50} %
YOLOv5n	2.51	7.1	38.4	75.8	65.2	73.5
YOLOv8n	3.15	8.8	42.7	76.5	66.8	74.9
YOLOv11n	2.58	6.4	38.9	77.2	68.5	77.1
YOLOv12n	2.56	6.4	37.2	78.3	70.2	77.6
RS-YOLO	**2.52**	**5.9**	**35.8**	**80.1**	**72.1**	**78.9**

In terms of accuracy, RS-YOLO achieves a Precision-value of 80.1%, an Recall-value of 72.1%, and an mAP_{50} of 78.9%, all of which far exceed those of other YOLO series algorithms, demonstrating superior precision. As shown in Table III, RS-YOLO performs exceptionally well in multi-category detection. For categories like “plane” and “tennis court,” the P-value exceeds 95%, the R-value exceeds 80%, and the mAP_{50} exceeds 90%; the mAP_{50-95} for “tennis court” is as high as 91.2%. Even for harder-to-detect categories like “bridge,” it maintains a certain level of accuracy. Overall, RS-YOLO balances efficiency and high accuracy, delivering outstanding performance across various scenarios.

Table IV presents the ablation experiment results of the RS-YOLO model proposed in this paper. The data shows that as the A2DyT and Inner-IoU modules are gradually introduced, the model's performance metrics continue to improve: Precision-value increased from 78.3% to 80.1%, recall-value improved from 70.2% to 72.1%, and mAP_{50} and mAP_{50-95} were optimized from 77.6% and 59.8% to 78.9% and 61.2%, respectively. Especially when the two modules work together, all metrics reach their optimal values, fully demonstrating the effectiveness of the A2DyT and Inner-IoU modules in enhancing the model's detection capabilities and highlighting their integrated value.

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TABLE III. IMPROVED RS-YOLO RESULTS FOR VARIOUS CATEGORIES.

Category	Images	Instances	Precision %	Recall %	mAP_{50} %	mAP_{50-95} %
plane	673	5,298	95.1	81.4	**93.8**	80.6
ship	640	19,571	92.0	84.8	**90.7**	74.2
storage tank	286	5,027	93.6	66.5	**84.5**	68.1
baseball-diamond	214	403	80.6	78.7	**81.2**	59.9
tennis court	247	1,602	96.6	88.3	**95.2**	91.2
basketball court	106	297	73.7	69.0	74.2	65.8
ground track field	263	316	68.7	55.4	63.2	49.6
harbor	660	4,789	89.3	78.5	**87.4**	58.7
bridge	456	897	66.4	53.9	60.3	36.3
large vehicle	456	9,257	90.7	81.6	**89.5**	71.8
small vehicle	448	10,833	61.8	73.6	72.7	56.1
helicopter	46	134	78.3	74.6	**82.4**	95.6
roundabout	225	299	72.6	62.8	67.0	50.8
soccer ball field	233	351	70.7	53.6	64.2	50.4
swimming pool	130	763	72.0	79.6	77.3	48.3

TABLE IV. IMPACT OF DIFFERENT MODULES ON NETWORK PERFORMANCE

YOLOv12	A2DyT	Inner-IoU	Precision %	Recall %	mAP_{50} %	mAP_{50-95} %
✓			78.3	70.2	77.6	59.8
✓	✓		79.5	71.3	78.4	60.5
✓		✓	78.9	70.8	77.9	60.1
✓	✓	✓	**80.1**	**72.1**	**78.9**	**61.2**

Fig.4 shows the visualization results of the RS-YOLO model for detecting multi-category remote sensing targets. The figure includes eight types of objects: ports, helicopters, airplanes, tennis courts, ships, bridges, storage tanks, and swimming pools. Fig.5 shows the performance of the RS-YOLO model over 300 training epochs, including changes in metrics such as box_loss, cls_loss, dfl_loss, accuracy, recall rate, average precision (mAP_{50}), and mAP_{50-95} during training and validation. These results indicate that the RS-YOLO algorithm demonstrates high accuracy and robustness when performing rotation change detection tasks in remote sensing images.

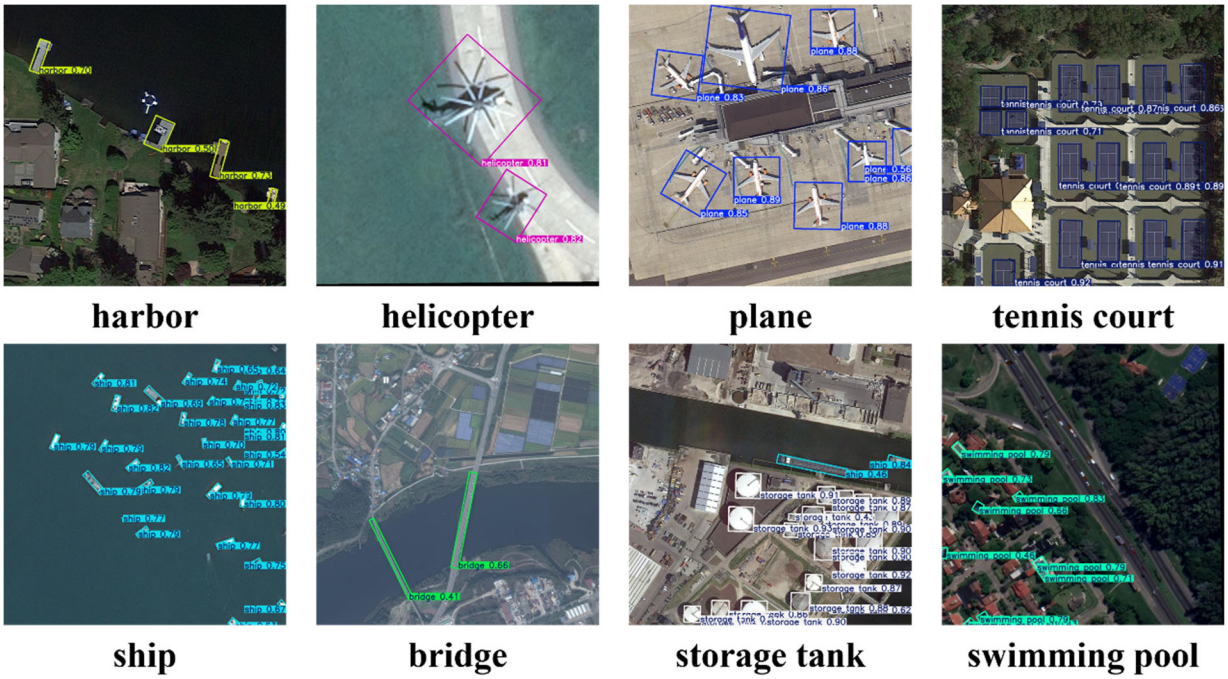


Fig. 4. RS-YOLO model detection results for multi-category remote sensing targets in the DOTA dataset

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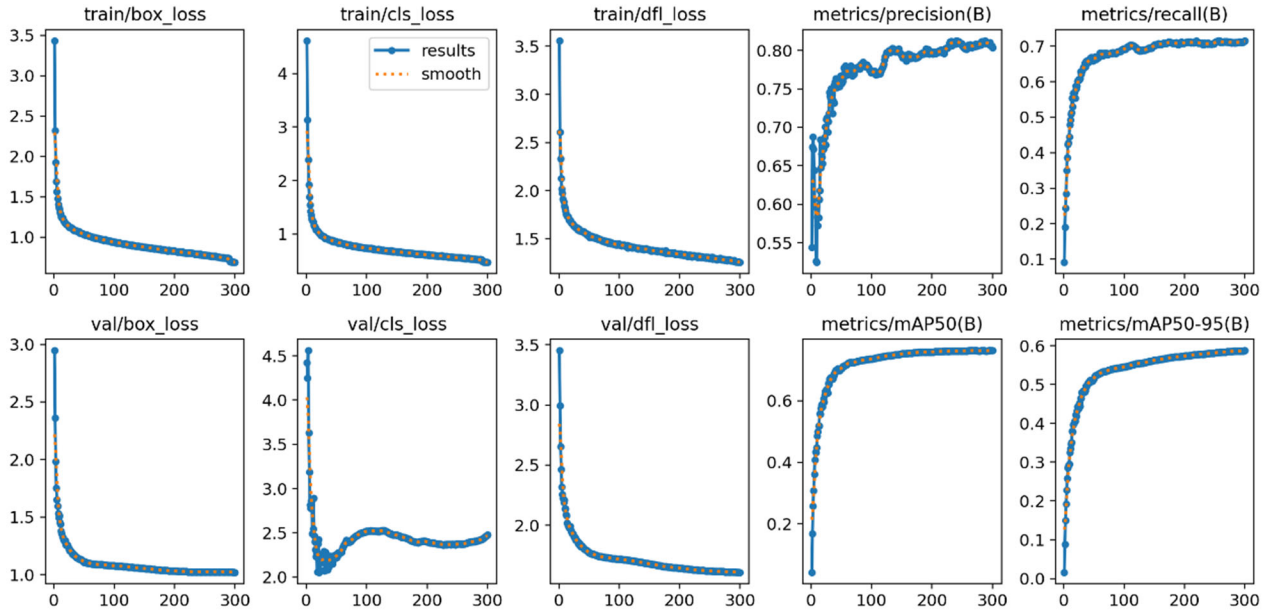


Fig. 5. Training curves of RS-YOLO over 300 epochs (including box loss, cls loss, mAP_{50}, etc.)

IV. Conclusion

This paper is based on the YOLOv12 object detection algorithm and has been extensively optimized and improved to address the characteristics of remote sensing images, resulting in the new method RS-YOLO. Compared to the original object detection algorithm, RS-YOLO demonstrates significant improvements in detection efficiency and accuracy. Through experiments on the DOTA dataset, RS-YOLO has demonstrated competitive detection performance and effectively improved detection accuracy. Comparative experiments with current mainstream object detection algorithms further demonstrate the applicability and robustness of RS-YOLO. While RS-YOLO has primarily been validated on the DOTA dataset, the application scenarios for remote sensing images extend far beyond this dataset. For very small or heavily occluded objects, we employ a series of local data augmentation strategies. This enhances the model's learning capabilities and detection performance in complex scenes, further improving its robustness. In the future, we can apply RS-YOLO to a wider range of remote sensing image datasets, such as high-resolution satellite images and drone aerial images, to explore its adaptability and optimization strategies in different scenarios.

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